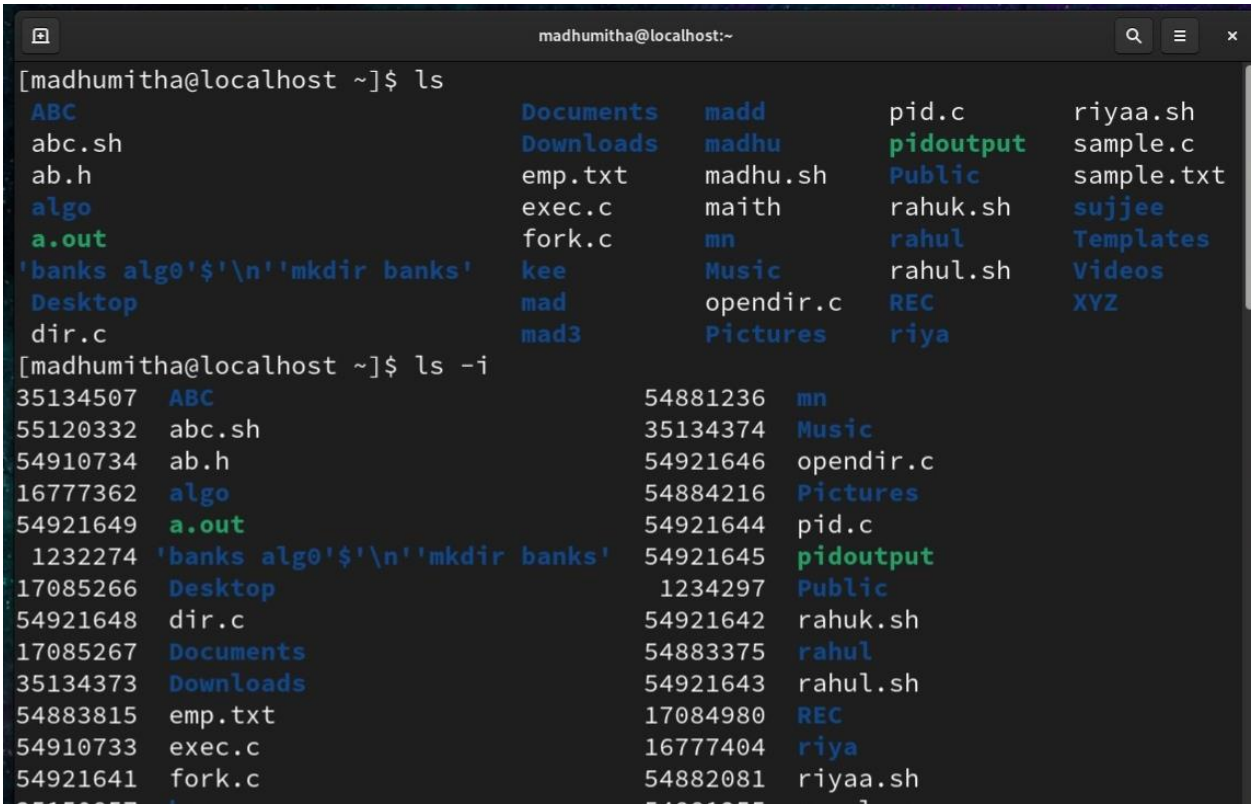
Operating system practical concepts

Basic linux commands

1. Ls commands

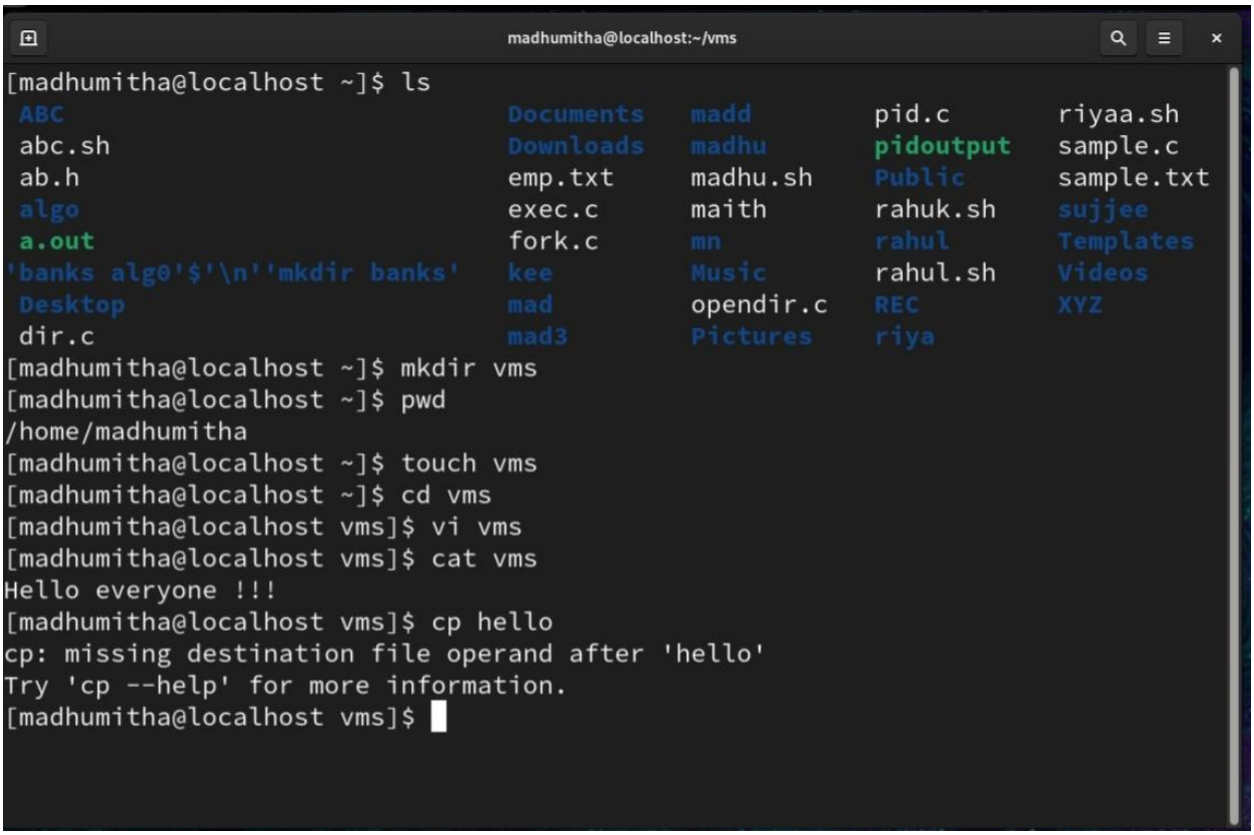


.

```
madhumitha@localhost:~
[madhumitha@localhost ~]$ ls -it
54921649  a.out                          54883806  sample.txt
54921646  opendir.c                      35150857  kee
54921648  dir.c                          54882068  XYZ
54921645  pidoutput                      35150850  mad3
54921644  pid.c                          55120332  abc.sh
54910733  exec.c                         54881255  sample.c
54921641  fork.c                         16777360  madd
54882081  riyaa.sh                        1232710  mad
54921643  rahul.sh                       35134507  ABC
54921642  rahuk.sh                       17084979  madhu
54883375  rahul                          17084980  REC
54881234  sujjee                         54881236  mn
16777404  riya                           17085266  Desktop
54881254  maith                          17085267  Documents
54884216  Pictures                       35134373  Downloads
16777362  algo                           35134374  Music
 1232274 'banks alg0'$'\n''mkdir banks'   1234297  Public
54910734  ab.h                           54884215  Templates
55120377  madhu.sh                        1234299  Videos
54883815  emp.txt
[madhumitha@localhost ~]$ ls -irt
 1234299  Videos      55120377  madhu.sh
54884215  Templates   54910734  ab.h
```

```
madhumitha@localhost:~
55120377  madhu.sh                          1234299  Videos
54883815  emp.txt
[madhumitha@localhost ~]$ ls -irt
 1234299  Videos      55120377  madhu.sh
54884215  Templates   54910734  ab.h
 1234297  Public       1232274 'banks alg0'$'\n''mkdir banks'
35134374  Music       16777362  algo
35134373  Downloads   54884216  Pictures
17085267  Documents   54881254  maith
17085266  Desktop     16777404  riya
54881236  mn          54881234  sujjee
17084980  REC         54883375  rahul
17084979  madhu       54921642  rahuk.sh
35134507  ABC         54921643  rahul.sh
 1232710  mad         54882081  riyaa.sh
16777360  madd        54921641  fork.c
54881255  sample.c    54910733  exec.c
55120332  abc.sh      54921644  pid.c
35150850  mad3        54921645  pidoutput
54882068  XYZ         54921648  dir.c
35150857  kee         54921646  opendir.c
54883806  sample.txt  54921649  a.out
54883815  emp.txt
[madhumitha@localhost ~]$
```

2. vi , pwd , touch ,cat commands



3. Banker's algorithm

```
madhumitha@localhost:~
[madhumitha@localhost ~]$ ls
 ABC                          Downloads   madhu.sh    pidoutput   sample.txt
 abc.sh                       emp.txt     maith       Public      sjf
 ab.h                         exec.c      mn          rahuk.sh    sujjee
 algo                         fork.c      Music       rahul       Templates
 a.out                        kee         opendir.c   rahul.sh    Videos
'banks alg0'$'\n''mkdir banks' mad        os_lab      REC         vms
 Desktop                      mad3        os_lab.c    riya        XYZ
 dir.c                        madd        Pictures    riyaa.sh
 Documents                    madhu       pid.c       sample.c
[madhumitha@localhost ~]$ vi os_lab
[madhumitha@localhost ~]$ gcc os_lab.c -o os_lab
[madhumitha@localhost ~]$ vi os_lab
Enter number of processes: 5
Enter number of resource types: 3
Enter Allocation Matrix:
0 1 0
2 0 0
3 0 2
2 1 1
0 0 2
Enter Max Matrix:
7 5 3
3 2 2
9 0 2
2 2 3
Enter Available Resources:
3 3 2
Safe sequence is: P1 P3 P4 P0 P2
[madhumitha@localhost ~]$
```

```c
include <stdio.h>

int main() {
    int n, r, i, j, k;

    // Input number of processes and resources
    printf("Enter number of processes: ");
    scanf("%d", &n);

    printf("Enter number of resource types: ");
    scanf("%d", &r);

    // int alloc(n][r], max[n][r], need[n][r], avail[r];
    int finish[n], safe_sequence[n];
    int count = 0, safeFound;

    // Allocation Matrix
    printf("Enter Allocation Matrix:\n");
     for(i = 0; i < n; i++) {
         for(j = 0; j < r; j++) {
             scanf("%d", &alloc[i][j]);
         }
    }

    // Allocation Matrix
    printf("Enter Allocation  Matrix:\n");
     for(i = 0; i < n; i++) {
         for(j = 0; j < r; j++) {
             scanf("%d", &max[i][j]));
    }

    // Max Matrix
    printf("Enter Max Matrix:\n");
     for(i = 0; i < n; i++) {
         for(j = 0; j < r; j++) {
             scanf("%d", &max[i][j]));
    }

    // Available Resources
    printf("Enter Available Resources:\n");
     for(i = 0; i < r; i++) {
         scanf("%d", &avail[i]);
    }
}
```

```
            int safe = 1;

            for(j = 0; j < r; j++) {
                if(need[i][j] > avail[j]) {
                    safe = 0;
                    break;
                }
            }

            if(safe) {
                for(k = 0; k < r; k++) {
                    avail[k] += alloc[i][k];
                }
            }

            safe_sequence[count] = i;
            finish[i] = 1;
            safeFound = 1;
            count++;
        }
    }

    if(!safeFound) {
        printf("System is in unsafe state! May occur deadlock.\n");
        return 0;
    }
}

// Print Safe Sequence
printf("Safe sequence is: ");
for(i = 0; i < n; i++) {
    printf("%d ", safe_sequence[i]));
}

return 0;

// Print Safe Sequence
printf("Safe sequence is: ");
for(i = 0; i < n; i++) {
    printf("%d ", safe_sequence[i]);
}
```

SCHEDULING ALGORITHMS :

1. Fcfs :

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1. Fcfs :

```
madhumitha@localhost:~
[madhumitha@localhost ~]$ ls
 ABC                          Downloads   madhu.sh    pidoutput   sample.txt
 abc.sh                       emp.txt     maith       Public      sjf
 ab.h                         exec.c      mn          rahuk.sh    sujjee
 algo                         fork.c      Music       rahul       Templates
 a.out                        kee         opendir.c   rahul.sh    Videos
'banks alg0'$'\n''mkdir banks' mad        os_lab      REC         vms
 Desktop                      mad3        os_lab.c    riya        XYZ
 dir.c                        madd        Pictures    riyaa.sh
 Documents                    madhu       pid.c       sample.c
[madhumitha@localhost ~]$ vi os_lab1.c
[madhumitha@localhost ~]$ gcc os_lab1.c -o os_lab1
[madhumitha@localhost ~]$ ./os_lab1
Enter number of processes: 3
Enter Arrival Time and Burst Time for P1: 0 5
Enter Arrival Time and Burst Time for P2: 1 3
Enter Arrival Time and Burst Time for P3: 2 8

Average Turnaround Time: 8.67
Average Waiting Time: 3.33
[madhumitha@localhost ~]$
```

madhumitha@localhost:~ — /usr/bin/vim os_lab1.c

```
[madhumitha@localhost ~]$ ls
-   int main() {
-       int n, r, j; j;
-
-       printf("Enter number of processes: ");
-       scanf("%d", &n);
-
-       int at[n], bt[n], ct[n], tat[n], wt[n];
-
-       // Input
-       for (int i =0; i < n; i++) {
-
-           printf("Enter Arrival Time and Burst Time for P%d: ", i+1);
-           scanf("%d %d", &at[i], &bt[i]);
-       }
-
-       float total_tat = 0, total_wt = 0;
-
-       // First process
-       ct[0] = at[0] + bt[0'];
-
-       // Remaining processes
-       for (int i = 1; < n;  i++) {
-           if(cct[i-1] < at[i])
-               ct[i] = at[i] -+ bt[i];  // CPU Idle
-           else
-               ct[i] = ct[i-1] + bt[i];
-       }
-
-       // Calculate TAT and WT
-       for (int i = 0; < n:++) {
-           tat[i] = ct[i] - at[i];
-           wt[i] = tat[i] - bt[i];
-
-       printf("\nAverage Turnaround Time: %22f", total_tat / n);
-       printf("\nAverage Waiting Time: %.2f\n", total_wt / n);
-   }  return 0;
```

2. sjf :

```
madhumitha@localhost:~
[madhumitha@localhost ~]$ ls
 ABC                          Downloads   madhu.sh    pid.c       sample.c
 abc.sh                       emp.txt     maith       pidoutput   sample.txt
 ab.h                         exec.c      mn          Public      sjf
 algo                         fork.c      Music       rahuk.sh    sujjee
 a.out                        kee         opendir.c   rahul       Templates
'banks alg0'$'\n''mkdir banks' mad        os_lab      rahul.sh    Videos
 Desktop                      mad3        os_lab1.c   REC         vms
 dir.c                        madd        os_lab.c    riya        XYZ
 Documents                    madhu       Pictures    riyaa.sh
[madhumitha@localhost ~]$ vi sjf.c
[madhumitha@localhost ~]$ vi sjf.c
[madhumitha@localhost ~]$ gcc sjf.c -o sjf
[madhumitha@localhost ~]$ ./sjf
Enter number of processes: 3
Enter Arrival Time and Burst Time for P1: 0 7
Enter Arrival Time and Burst Time for P2: 2 4
Enter Arrival Time and Burst Time for P3: 4 1

Average Turnaround Time: 7.00
Average Waiting Time: 3.00

[madhumitha@localhost ~]$
```

```c
int main()
{
    int n;
    printf("Enter number of processes: ");
    scanf("%d", &n);

    int at[n], bt[n], ct[n], tat[n], wt[n];

    int done[n] = 0;

    int completed = 0;
    int completionTime = 0;
    float avgtat =0, avgwt = 0;

    // SJF Scheduling (Non-Preemptive)
    while(completed < n)
    {
       int idx = -1;
       int minBT = 9999;

       for(int i = 0; i < n;  i++)
       {
           if(at[i] <= completionTime && done[i] == 0)
           {
               if(bt[i] < minBT)
               {
                   minBT = bt[i];
                   idx = i;
               }
           }
       }
    }
}
```

```
[madhumitha@localhost~]                                    1,1
```

Round robin

```
madhumitha@localhost:~
[madhumitha@localhost ~]$ ls
 ABC                          Downloads   madhu.sh    pid.c       sample.c
 abc.sh                       emp.txt     maith       pidoutput   sample.txt
 ab.h                         exec.c      mn          Public      sjf.c
 algo                         fork.c      Music       rahuk.sh    sujjee
 a.out                        kee         opendir.c   rahul       Templates
'banks alg0'$'\n''mkdir banks' mad        os_lab      rahul.sh    Videos
 Desktop                      mad3        os_lab1.c   REC         vms
 dir.c                        madd        os_lab.c    riya        XYZ
 Documents                    madhu       Pictures    riyaa.sh
[madhumitha@localhost ~]$ vi roundrobin.c
[madhumitha@localhost ~]$ gcc roundrobin.c -o roundrobin
[madhumitha@localhost ~]$ ./roundrobin
bash: ./roundrobin: No such file or directory
[madhumitha@localhost ~]$ ./roundrobin
Enter number of processes: 3
Enter Time Quantum: 2
Enter AT and BT for P1:  0 5
Enter AT and BT for P2:  1 3
Enter AT and BT for P3:  2 2

Average Turnaround Time: 6.33
Average Waiting Time: 3.33
[madhumitha@localhost ~]$
```

madhumitha@localhost:~ — /usr/bin/vim roundrobin.c

```c
#nclude <stdio.h>
int main()
{
    int n, tq, completed = 0, completionTime = 0;

    printf("Enter number of processes: \n");
    scanf("%d", &n));

    printf("Enter Time Quantum: \n");
    scanf("%d", &tq);

    irt at[n], bt[n], rem[n], tat[n], wt[n];
    float avgtat = 0, avgwt = 0;

    // Tnput
    for (int i = 0; i < n; i++)
    {
        printf("Enter AT and BT for P%d: '', i + 1);
        scanf("%d %od, &at[i], &bt[i]);
        rem[i] = bt[i];   // Remaining time = Burst time

    // Round Robin Logic
    while (completed < n)
    {
        int idle = 1;

        for (int i = 0; i < n; i++)
        {
            if (at[i] <= completionTime && rem[i] > 0)
            {
                idle = 0;
```

madhumitha@localhost:~ — /usr/bin/vim roundrobin.c

```c
    int idle = 1;

    for (int i = 0; i < n; i++)
    {
        if (at[i] <= completionTime && rem[i] > 0)
        {
            idle = 0;

            if (rem[i] > tq)
            {
                completionTime += tq;
                rem[i] == tq;
            }
            else
            {
                completionTime += rem[i];
                tat[i] = completionTime - at[i];
                wt[i] = tat[i] - bt[i];
                rem[i] = 0;
                completed++;
            }
        }
    }
    if (idle)
         completionTime ++;
}

// Calculate averages
for (int i = 0; i < n; i++)
{
     avgat += tat[i];
     avgwt += wt[i];
}

printf("Average Turnaround Time: %.2f\n", avgat / n);
printf("Average Waiting Time: %.2f\n", avgwt / n);
return 0;
```

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Priority scheduling

```
madhumitha@localhost:~
[madhumitha@localhost ~]$ ls
-       banker.c     Documents     mait      Music          Pictures        sh              whilw.sh
a       bankers      Downloads     maitre    os_lab1        Public          sjf             XYZ
ABC     bm           h             mash      os_lab1.c      REC             sjf.c
abc.sh  close        mai           mithii    os_lab1.c      sample.c        tesatstxt
abc.sh  Desktop      maithu        os_lab1   roundrobin.c   sample.txt      whilw.sh
shf     Documents    maitre        os_lab1.c XYZ
[madhumitha@localhost ~]$ vi priority
[madhumitha@localhost ~]$ gcc priority.c o priority
[madhumitha@localhost ~]$ ./priority
Enter number of processes: 3

Enter AT, BT and Priority for P1: 0 5 2

Enter AT, BT and Priority for P2: 1 3 1

Enter AT, BT and Priority for P3: 2 8 3
2 8 3

Average Turnaround Time: 8.67
Average Waiting Time: 3.33
[madhumitha@localhost ~]$
```

madhumitha@localhost:~ — /usr/bin/vim priority.c

```c
include <stdio.h>

int main()
{
    int n, completed = 0, time = 0;

    printf("Enter number of processes: \n");
    scanf("%d", &n);

    int at[n], bt[n], priority[n], done[n]);
    int at[n], tat[n], wt[n];
    float avgTat = 0, avgwt = 0;

    // Input
    for (int i = 0; i < n; i++)
    {
        printf("Enter AT, BT and Priority for P%d: \n", i + 1);
        scanf("%d %d %d", &at[i], &bt[i], &priority[i]);
        done[i] = 0;
    }

    // Scheduling Logic (Non-Preemptive Priority: lower number = higher priority)
    while (completed < n)
    {
        int idx = -1;
        int highPriority = 9999;

        for (int i = 0; i < n; i++)
        {
            if (at[i] <= time && done[i] == 0)
            {
                if (priority[i] < highPriority)
                {
                    highPriority = priority[i];
                    idx = 1;
                }
            }
        ]   if (idx != -1)
```

1,1

madhumitha@localhost.~ —/usr/bin/vim priority.c

```
    if (idx != -1)
    {
        time += bt[idx];
        bt[idx] = time;
        tat[idx] = at[idx] = at[idx];
        wtt[idx] = bt[idx] = bt[idx];

        done[idx] = 1;
        completed++;
    }
    else
    {
        time++;   // CPU Idle
    }
}

// Average calculation
for (int i = 0; < < n; i++)
{
    avgTat += tat[i];
    avgWt += wt[i];
}

printf("Average Turnaround Time: %.2f\n", avgTat / n);
printf("Average Waiting Time: %.2f\n", avgWt / n);

return 0;
```